

# Interval-Certified Positivity of a Corrected GLSM Metric on a Six-Dimensional Product Neighborhood of the Calabi–Yau $(5, 81)$

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## Abstract

We give a computer-assisted proof of positive definiteness for a frozen corrected gauged-linear-sigma-model (GLSM) quotient metric on a six-real-dimensional product neighborhood in one explicit toric chart of a compact Calabi–Yau threefold with  $(h^{2,1}, h^{1,1}) = (5, 81)$ . This consolidates the central-path calculation, its independent endpoint validation, and the full product-tube extension into one theorem hierarchy. The parameter domain is

$$\mathcal{T} = [-3/20000, 0] \times [-10^{-8}, 10^{-8}]^5.$$

A parameterized interval Krawczyk calculation certifies a locally unique 83-real-dimensional hypersurface–moment-map root on 1,084 overlapping parent boxes. A robust central continuation branch anchors these local roots, and 1,083 nonempty product overlaps glue them into one branch on  $\mathcal{T}$ . The corrected  $3 \times 3$  Hermitian pullback metric is reconstructed on 34,688 child boxes that exactly tile the transverse cube. All 104,064 directed rational intervals for its three leading principal minors have positive lower bounds. The smallest third-minor lower bound is

$$\frac{2100827914296867}{18889465931478580854784} > 0.$$

At the endpoint, a separate 83-real Krawczyk implementation and independent baseline/correction matrix reconstruction reprove local positivity. On a fixed ensemble of 31 Gaussian-integer directions, two directions have a certified relative correction above one percent; the largest lower bound is  $0.0183143620\dots$ . The release separates a dependency-free exact last-mile verifier from a full Arb formula-regeneration layer, validates integer geometry without rounding, and includes fail-closed mutation tests. The theorem is local to one chart and one frozen Kähler ansatz; it is neither a global Calabi–Yau positivity theorem nor a Ricci-flatness or physical-vacuum certificate.

## 1 Introduction

Yau’s theorem gives a unique Ricci-flat Kähler metric in a fixed Kähler class after normalization, but explicit compact Calabi–Yau metrics are rarely available. Numerical approaches based on balanced metrics, energy functionals, and machine learning have made quantitative calculations possible on increasingly complicated geometries [1, 2, 3, 4, 5, 6, 7]. Their usual validation language is statistical or residual based. It is different from a uniform theorem in which every rounding and truncation error is enclosed on a continuum of parameter values.

The present work develops the latter type of statement for an explicit corrected GLSM quotient metric. The geometry is the  $(5, 81)$  threefold used in the type-IIB compactification

data of Ref. [8]. That provenance makes the example identifiable and non-toy, but it does not transfer the physical conclusions of that compactification to our local metric calculation. Related numerical work has studied leading metric corrections on compact Calabi–Yau spaces [9]; rigorous interval methods have also appeared in neighboring Calabi–Yau computations, for example in monodromy calculations [10]. We make no bibliometric priority claim here. The mathematical contribution is the explicit theorem, its complete box cover, and a release that permits exact replay of the final inequalities.

The calculation was developed in two stages that are consolidated here. The first stage certified positivity along the central one-dimensional continuation path and supplied an independently reconstructed endpoint root, metric, and nonzero-correction witness. A path theorem alone does not imply an open-neighborhood theorem: transverse variables can alter both the implicit root and the metric, and a finite collection of sampled corners is not a cover. The second stage proves the complete product tube. Root existence and uniqueness are established on parents of transverse radius  $10^{-8}$ ; metric positivity is then proved on their  $2^5$  exact transverse children. Earlier versioned Zenodo/GitHub deposits are computational prereleases of this unified manuscript, not separate peer-reviewed articles.

The distinction between discovery and proof is essential. Correction coefficients were obtained in an approximation stage and subsequently frozen as exact binary rationals. Their provenance does not prove positivity. Positivity follows from interval inclusion for the implicit system and strict rational lower bounds for the three principal minors. Likewise, floating-point contraction ratios retained for diagnostics are never proof inputs.

## 2 Geometry and frozen corrected metric

Let  $Q \in \mathbb{Z}^{81 \times 85}$  denote the disclosed GLSM charge matrix. In the selected regular toric chart, let  $z_i$  be Cox coordinates and solve the moment-map equations

$$QD = r, \quad D_i = |z_i|^2 \exp((Q^T u)_i), \quad i = 1, \dots, 85, \quad (1)$$

for  $u \in \mathbb{R}^{81}$ . Writing  $D = \text{diag}(D_i)$ , the horizontal ambient quotient tensor is

$$H = D - DQ^T(QDQ^T)^{-1}QD. \quad (2)$$

If  $A$  is the analytic tangent matrix obtained by implicit differentiation of the hypersurface equation, the baseline pullback metric is evaluated in the numerically more convenient form

$$G_0 = A^*DA - (QDA)^*(QDQ^T)^{-1}(QDA). \quad (3)$$

The correction is the complex Hessian of a frozen scalar. Put

$$S = \sum_{i=1}^{85} D_i, \quad p_i = D_i/S, \quad (4)$$

and write  $S_0, p_i^{(0)}$  for their values at the disclosed reference point. For an archived rational direction  $v \in \mathbb{Q}^{85}$ , define

$$q = \frac{S - S_0}{S + S_0}, \quad s = \sum_{i=1}^{85} v_i (p_i - p_i^{(0)})^2. \quad (5)$$

The potential and corrected metric are

$$\phi = c_1(\lambda_1 q^2)^2 + c_2(\lambda_2 s)^2, \quad G_{\text{corr}} = G_0 + \partial \bar{\partial} \phi. \quad (6)$$

All entries of  $v$ , all reference data, and  $c_1, c_2, \lambda_1, \lambda_2$  are stored as reduced rational pairs. In particular,

$$c_1 = -\frac{3704684168964555}{9007199254740992}, \quad c_2 = \frac{4163219474253507}{1125899906842624}. \quad (7)$$

The scalar is invariant under the complexified GLSM action, so its  $\partial\bar{\partial}$  deformation is a well-defined local Kähler-form candidate on the regular quotient chart. The task below is to prove positive definiteness on a declared domain, not to prove Ricci flatness.

The archived adapter historically represented the entries of  $Q$  and the hypersurface exponent vectors as JSON numbers, some spelled as integral binary64 values such as 1.0. The publication implementation accepts such a value only if it is exactly integral and has magnitude at most  $2^{53}$ . There is no call to `round`. The resulting  $81 \times 85$  charge matrix, ten 85-component exponent vectors, coefficients, and chart indices are serialized in a separate exact-integer registry and checked against the raw inputs before verification.

### 3 The product domain and implicit system

Let  $d$  be the affine continuation coordinate and set  $t = d - 3/2$ . Five real transverse chart coordinates are denoted by  $w = (w_1, \dots, w_5)$ . The certified parameter tube is

$$\mathcal{T} = \left[ -\frac{3}{20000}, 0 \right] \times \left[ -\frac{1}{10^8}, \frac{1}{10^8} \right]^5. \quad (8)$$

The coupled real system consists of the real and imaginary hypersurface equations and 81 moment-map equations:

$$F(t, w, x) = 0, \quad x \in \mathbb{R}^{83}. \quad (9)$$

For a parameter box  $P$ , a center  $x_0$ , an unknown box  $X = x_0 + [-r, r]$  (with componentwise radii), and a midpoint inverse  $C$ , the parameterized Krawczyk operator is

$$K(P, X) = x_0 - CF(P, x_0) + (I - CF_x(P, X))(X - x_0). \quad (10)$$

Strict componentwise inclusion  $K(P, X) \subset \text{int } X$  proves existence of a root for every parameter in  $P$  and uniqueness among roots in  $X$ ; regularity follows from the same enclosure [11, 12]. The complex hypersurface elimination additionally stores a directed interval for its derivative and checks that the complex rectangle excludes zero.

The longitudinal centers form an exact ordered grid of 1,084 values. Every parent has longitudinal half-width  $7 \times 10^{-8}$  and transverse half-width  $10^{-8}$  in each of the five directions, with clipping only when interpreting the union at the two declared path endpoints. The smallest overlap width between consecutive longitudinal parents is

$$\frac{27}{18050000000} > 0. \quad (11)$$

For every parent the calculation stores two hypersurface and 81 D-term image intervals. Thus the full root layer contains  $1084 \times 83 = 89972$  exact componentwise strict inclusions. Its smallest distance from an image endpoint to the corresponding unknown-box boundary is

$$\frac{9335426793689657913651}{10910555522022028301723238400000} > 0. \quad (12)$$

## 4 Gluing the local roots

The Krawczyk theorem is local in the unknown variables. A separate argument is therefore needed before the 1,084 local maps may be called one branch. The anchor is the previously certified robust continuation chain on the central slice  $w = 0$ . It consists of seven Krawczyk patches and seven bridges, covers  $t \in [-3/20000, 0]$ , and has exact maximum image-to-radius ratios

$$\frac{4808980629275713}{5661108081689003} < \frac{17}{20}, \quad \frac{5163879392018433}{6226644134068241} < \frac{17}{20} \quad (13)$$

for patches and bridges, respectively. All  $7 \times 83$  patch and  $7 \times 83$  bridge component inequalities are replayed by the portable verifier.

**Lemma 1** (Product branch gluing). *Let  $P_i$  be the ordered parent parameter boxes and  $X_i$  their Krawczyk uniqueness boxes. Suppose: (i) each  $P_i$  has a continuous locally unique root map  $x_i : P_i \rightarrow X_i$ ; (ii) at the center of the central slice  $w = 0$ , the robust continuation root is strictly nested in  $X_i$ ; and (iii) every  $P_i \cap P_{i+1}$  is nonempty and connected. Then the maps  $x_i$  glue to one continuous locally unique root branch on  $\bigcup_i P_i$ .*

*Proof.* On the connected central slice of  $P_i$ , consider the set where  $x_i$  equals the robust continuation root. It is nonempty by the strict anchor nesting, closed by continuity, and open by local uniqueness and strict interior inclusion. Hence the two maps agree throughout that slice. On the connected product overlap  $P_i \cap P_{i+1}$ , the analogous equality set for  $x_i$  and  $x_{i+1}$  contains the nonempty central slice. It is again open and closed, so the two local maps agree on the overlap. Induction over the 1,083 edges gives the claimed branch.  $\square$

**Proposition 1** (Certified implicit branch). *For every  $(t, w) \in \mathcal{T}$ , Eq. (9) has a root on the branch selected by the robust central anchor. The root is locally unique in the union of the declared Krawczyk boxes and depends continuously on  $(t, w)$ .*

This proposition is deliberately local in  $x$ : it does not exclude unrelated roots outside the certified boxes.

## 5 Metric cover and positivity

The root parent at a fixed longitudinal center serves 32 metric children. For each transverse coordinate the two child intervals are

$$[-10^{-8}, 0] \quad \text{and} \quad [0, 10^{-8}], \quad (14)$$

with centers  $\mp 5 \times 10^{-9}$  and half-width  $5 \times 10^{-9}$ . Their Cartesian products give exactly  $2^5 = 32$  closed boxes. Endpoint comparison proves that their union is the parent transverse cube without a gap. Across all longitudinal centers there are

$$1084 \times 32 = 34688 \quad (15)$$

metric boxes.

On every child, interval implicit differentiation supplies the six real first and second derivatives required for the  $3 \times 3$  Hermitian pullback tensor. The inverse of  $QDQ^T$  and the joint implicit Jacobian is enclosed using a scaled finite Neumann series with a geometric remainder. Second-order interval jets analytically differentiate Eq. (6). The Hermitian tensor is finally intersected with its conjugate transpose.

Let  $\Delta_k$  be the determinant of the leading  $k \times k$  principal submatrix of  $G_{\text{corr}}$ . The Hermitian Sylvester criterion gives

$$G_{\text{corr}} \succ 0 \iff \Delta_1 > 0, \quad \Delta_2 > 0, \quad \Delta_3 > 0. \quad (16)$$

All three directed intervals are stored on every child. Table 1 lists the exact global minima of their lower endpoints. The maximum ratio of a robust-anchor radius to its containing parent radius is

$$\frac{146013824139625356625}{244296187218972043640832} < 1. \quad (17)$$

| Certified quantity                        | Exact value                                  |
|-------------------------------------------|----------------------------------------------|
| Parameter domain                          | $[-3/20000, 0] \times [-10^{-8}, 10^{-8}]^5$ |
| Root parents / metric children            | 1084/34688                                   |
| Root component inclusions                 | 89972                                        |
| Longitudinal gluing edges                 | 1083                                         |
| $\min \underline{\Delta}_1$               | 3459225413190877/288230376151711744          |
| $\min \underline{\Delta}_2$               | 6861360338265545/18446744073709551616        |
| $\min \underline{\Delta}_3$               | 2100827914296867/18889465931478580854784     |
| Minimum root inclusion margin             | Eq. (12)                                     |
| Minimum longitudinal overlap              | 27/18050000000                               |
| Maximum robust-anchor/parent-radius ratio | Eq. (17)                                     |

Table 1: Exact results recomputed by the dependency-free verifier.

**Theorem 1** (Corrected-metric positivity on the certified product tube). *In the selected GLSM chart, for the exactly specified frozen ansatz of Eq. (6), there is a locally unique parameterized root branch on  $\mathcal{T}$ . Along that branch the corrected Hermitian pullback metric is positive definite at every point of  $\mathcal{T}$ .*

*Proof.* The exact parent grid covers the longitudinal interval, and the 32 child templates tile each transverse cube. The 1,084 parametric Krawczyk inclusions give local root existence and uniqueness on all parents. Lemma 1 identifies these roots with one branch on their union. On each of the 34,688 metric children, interval evaluation gives positive rational lower bounds for all three leading principal minors. Their global minima are the positive fractions in Table 1. Equation (16) proves positive definiteness on every child and therefore on all of  $\mathcal{T}$ .  $\square$

**Proposition 2** (Central-path corollary). *On the central slice  $u_1 = \dots = u_5 = 0$ , the same locally unique branch exists for  $d \in [1.49985, 1.5]$ , and its corrected Hermitian pullback metric is positive definite throughout the interval.*

*Proof.* The slice is contained in  $\mathcal{T}$  under  $t = d - 1.5$ . The claim is therefore the restriction of Theorem 1. In particular, no sampling argument or additional continuity assumption is used.  $\square$

## 6 Independent endpoint validation and correction materiality

The product-tube proof already contains the endpoint  $d = 1.5$ . As a cross-check, the release also preserves the independently implemented endpoint calculation from the first development stage. It starts from a separate 83-real root enclosure centred at  $d = 3/2$ , with free radius  $10^{-9}$ , and does not reuse the primary tube root images. Its exact Krawczyk replay gives a maximum root-image ratio

$$\frac{2151456907264000390625}{4722366482869645213696} = 0.4555887212 \dots < 1.$$

The baseline and correction matrices are independently reconstructed, and all determinants are evaluated by the exact Leibniz formula. The resulting lower bounds for the three leading

| Endpoint quantity | Certified lower bound             |
|-------------------|-----------------------------------|
| $\Delta_1$        | 0.0127447625596...                |
| $\Delta_2$        | 0.000415809951340...              |
| $\Delta_3$        | $1.46941475236... \times 10^{-6}$ |

Table 2: Independent endpoint reconstruction. Exact numerators and denominators are recomputed by the verifier.

principal minors are listed in Table 2. The exact rational pairs are proof inputs; decimals are printed here only for readability.

Positivity alone does not show that the correction is quantitatively visible. We therefore freeze a finite ensemble of 31 Gaussian-integer tangent directions and compare the correction quadratic form with the baseline form on the endpoint enclosure. Two directions have a rigorously positive relative correction exceeding one percent. The largest certified lower bound is

$$\frac{7568816278373741}{413272176438128128} = 0.0183143620836\dots \quad (18)$$

This is an existence statement within the disclosed finite ensemble. It is not an operator-norm bound, a uniform statement along the whole path, or a claim that the frozen correction is the complete physical  $\alpha'$  series. The independent endpoint calculation is not needed for Theorem 1; its value is implementation diversity and a separate exact consistency check at a shared point.

## 7 Proof architecture and reproducibility

The release has two deliberately different verification levels.

The *quick verifier* uses only the Python standard library. It checks canonical reduced rational pairs, the exact integer registry, all robust-chain inclusions, all 89,972 parent-root inclusions, strict derivative exclusion of zero, robust-anchor nesting, exact longitudinal coverage, all 1,083 overlap widths, the 32-box transverse tiling, and all 104,064 stored metric-minor intervals. It recomputes the minima in Table 1; serialized Boolean verdicts and floating-point ratios are not accepted as proof inputs. It also reconstructs the independent endpoint determinants and the fixed-direction materiality bounds from exact rational data.

The *full regeneration* uses python-flint 0.9.0 with FLINT 3.6.0 and 100 decimal digits of Arb working precision [13]. From the frozen formula inputs it reconstructs every parent root enclosure and every metric child enclosure, then compares all proof-relevant fields with the archived release. Diagnostic floating-point fields and runtime metadata are excluded from this equality test. The full calculation is resumable; its checkpoint directory is explicitly outside the immutable proof manifest. The release-level run completed all 1,084 root parents and all 34,688 metric children; every proof-relevant regenerated field equals its archived value.

For a published release, the commands are

```
python verify_release.py --quick
python verify_release.py --full
```

The first command is intended for routine reviewer replay. The second is the slower reconstruction audit. Eleven mutation tests force equality in a Krawczyk inclusion, put zero into a metric lower bound, delete a transverse child, corrupt the integer geometry, break a robust anchor, and alter endpoint evidence. Each mutation must be rejected. Finally, a SHA-256 manifest binds every release file byte for byte.

This is a computer-assisted proof, not a proof in a foundational proof assistant. The quick layer trusts Python integer arithmetic, JSON parsing, and SHA-256, as well as the

mathematical implication from valid directed Arb enclosures to their serialized rational supersets. The regeneration layer additionally trusts the disclosed formula implementation, Python, python-flint, FLINT/Arb, the compiler, and hardware. Separating the layers makes these assumptions visible and prevents a cached Boolean from substituting for an inequality.

## 8 Scope and interpretation

Theorem 1 is stronger than a path-only theorem because  $\mathcal{T}$  has nonzero extent in all five declared transverse directions. It is nevertheless local in several independent senses.

First, the theorem concerns one toric chart and one small product neighborhood. It does not cover the whole compact hypersurface or establish compatibility with other charts. “Positive on the (5, 81) Calabi–Yau” without this qualification would therefore be an overstatement.

Second,  $G_0$  is a section-free GLSM quotient reference metric, not a certified Ricci-flat Calabi–Yau metric. Equation (6) is one frozen gauge-invariant approximation ansatz. We prove positivity of that specific tensor. We do not bound a global Ricci or Monge–Ampère residual, derive a complete physical  $\alpha'$  correction, or control omitted orders.

Third, the implicit theorem establishes local uniqueness in disclosed boxes. It neither proves global uniqueness of the algebraic/moment-map system nor excludes disconnected branches elsewhere in field space.

Finally, the geometry originates in a realistic compactification program, but the present theorem says nothing by itself about the full 113-field system, moduli stabilization, masses, uplift, metastability, or phenomenology. It is not a certificate of a physical vacuum.

## 9 Conclusion

We have consolidated a one-dimensional positivity calculation and its independent endpoint audit into a uniform computer-assisted theorem on an explicit six-real-dimensional product tube. The former path theorem is now the immediate central-slice corollary, while the endpoint calculation supplies an implementation-diverse validation and a certified finite-ensemble witness that the correction can exceed one percent. The proof combines 1,084 parametric root parents, an anchored branch-gluing argument, an exact  $2^5$  transverse tiling, and 34,688 interval metric boxes. Every final inequality is available for dependency-free rational replay, while a separate Arb layer supports formula regeneration from frozen inputs.

The natural mathematical extensions are larger certified radii and overlapping multi-chart coverage. A physically stronger direction would require an independent approximation theorem relating the frozen ansatz to a canonical Ricci-flat or effective-action metric. Neither extension is assumed in the result reported here.

## Data and code availability

The versioned release contains the manuscript, compressed proof records, exact inputs, quick verifier, full Arb regeneration code, mutation tests, independent endpoint evidence, trust model, certificate schema, and SHA-256 manifest. Code is distributed under the BSD 3-Clause License. Earlier GitHub/Zenodo versions are retained as computational provenance; this combined release supersedes them as the manuscript intended for peer review. Repository and archival DOI metadata will be inserted only after the immutable public release is created.

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